

Counterexamples to Self-Adjoint/Hermitian Implications on Smooth Banach Spaces

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Source and Question

The source paper is Mohammed Shameem and Deepesh K. P., *On properties of normal operators and self-adjoint operators on smooth Banach spaces*, arXiv:2605.15898, submitted 15 May 2026. It studies operators on complex smooth Banach spaces using the normalized duality map J . The first item in its final list of open problems asks:

Under what conditions does a self-adjoint operator become Hermitian and conversely?

The same final list asks about numerical ranges of normal, unitary, and self-adjoint operators. The source also gives, on $\ell^4(\mathbb{N})$, the coordinate swap as an example of a normal, self-adjoint, and unitary operator.

Definitions

Let X be a complex smooth Banach space and let $J : X \rightarrow X^*$ be the normalized duality map, so that

$$\|Jx\| = \|x\|, \quad Jx(x) = \|x\|^2.$$

Following the source, an operator $T \in B(X)$ is

self-adjoint	if $T'J = JT$,
Hermitian	if $Jx(Tx) \in \mathbb{R}$ for all $x \in X$,
positive	if $Jx(Tx) \geq 0$ for all $x \in X$,
normal	if $\ Tx\ = \ T'Jx\ $ for all $x \in X$,
unitary	if $\ Tx\ = \ T'Jx\ = \ x\ $ for all $x \in X$.

On $X = \ell_4^2 = (\mathbb{C}^2, \|\cdot\|_4)$, the normalized duality functional is

$$J(z_1, z_2)(w_1, w_2) = \frac{|z_1|^2 \overline{z_1} w_1 + |z_2|^2 \overline{z_2} w_2}{\|(z_1, z_2)\|_4^2}.$$

Equivalently, under the usual identification of $(\ell_4^2)^*$ with $\ell_{4/3}^2$, the coefficient vector of Jz is

$$\frac{(|z_1|^2 \overline{z_1}, |z_2|^2 \overline{z_2})}{\|z\|_4^2}.$$

Proof Intuition

The Hilbert-space equivalence between self-adjointness and Hermitian numerical values uses conjugate symmetry of the inner product. The duality map of ℓ_4^2 gives only a semi-inner-product-like pairing, and it is not conjugate symmetric. A coordinate swap still commutes perfectly with the duality map, so it is self-adjoint and even unitary in the source's sense, but the value $Jx(Ux)$ can point in a genuinely imaginary direction. Conversely, a real diagonal multiplier has real nonnegative duality values, but the normalization in J prevents it from satisfying $T'J = JT$ unless the diagonal entries have the same modulus in the relevant sense.

Theorem 1. *On complex smooth Banach spaces, neither implication*

$$\text{self-adjoint} \Rightarrow \text{Hermitian}, \quad \text{Hermitian} \Rightarrow \text{self-adjoint}$$

holds in general. In fact both failures occur on $X = \ell_4^2$. Moreover, there is a normal, unitary, self-adjoint operator on X whose numerical range is not contained in the real line.

Proof. Let $U : \ell_4^2 \rightarrow \ell_4^2$ be the coordinate swap

$$U(z_1, z_2) = (z_2, z_1).$$

The transpose U' is the corresponding coordinate swap on $\ell_{4/3}^2$. Since the formula for J is symmetric in the two coordinates,

$$U'J(z_1, z_2) = J(z_2, z_1) = J(U(z_1, z_2)).$$

Thus U is self-adjoint. Also U is a surjective isometry, and the same coordinate calculation gives $\|U'Jx\|_{4/3} = \|x\|_4 = \|Ux\|_4$; hence U is unitary and normal in the source's definitions.

However, take $x = (1, 2i)$. Then $\|x\|_4^2 = \sqrt{17}$, and

$$\begin{aligned} Jx(Ux) &= \frac{|1|^2 \overline{1}(2i) + |2i|^2 \overline{2i}(1)}{\sqrt{17}} \\ &= \frac{2i + 4(-2i)}{\sqrt{17}} = -\frac{6i}{\sqrt{17}}, \end{aligned}$$

which is not real. Therefore U is self-adjoint, unitary, and normal, but not Hermitian. If $y = x/\|x\|_4$, then $y \in S_X$ and

$$Jy(Uy) = -\frac{6i}{17}.$$

Consequently the numerical range of this self-adjoint, unitary, normal operator is not contained in $\mathbb{R}$.

For the reverse implication, let $D : \ell_4^2 \rightarrow \ell_4^2$ be

$$D(z_1, z_2) = (z_1, 2z_2).$$

For every nonzero $z = (z_1, z_2)$,

$$Jz(Dz) = \frac{|z_1|^4 + 2|z_2|^4}{\|z\|_4^2} \in [0, \infty).$$

Thus D is Hermitian, and even positive. But D is not self-adjoint. Indeed, for $z = (1, 1)$,

$$D'Jz = \frac{(1, 2)}{\sqrt{2}}, \quad J(Dz) = J(1, 2) = \frac{(1, 8)}{\sqrt{17}},$$

as coefficient vectors in $\ell_{4/3}^2$. These two vectors are not equal, so $D'Jz \neq J(Dz)$, and D is not self-adjoint. $\square$

Verification Notes

No numerical optimization is used. The arithmetic check in `code/check_14_examples.py` evaluates the displayed duality-map values and the two coefficient vectors witnessing $D'Jz \neq J(Dz)$.

The counterexample is finite-dimensional, so X is reflexive, smooth, and strictly convex. It is therefore inside the regularity hypotheses used in the source paper's later unitary and spectral results.

Novelty and Scope

The run's cheap indexes were searched for arXiv:2605.15898, the source title, normal operators, self-adjoint operators, Hermitian operators, smooth Banach spaces, and related terms. No existing packet or attempt for this source was found. A bounded web search for the source title, García-Pacheco self-adjointness, Hermitian operators, smooth Banach spaces, and ℓ^4 found the source paper and background references, but no separate answer to the first open question.

This does not characterize the additional hypotheses under which either implication becomes true. It gives the baseline negative answer: without extra hypotheses, the two notions are incomparable, even in dimension two.

References

- Mohammed Shameem and Deepesh K. P., *On properties of normal operators and self-adjoint operators on smooth Banach spaces*, arXiv:2605.15898, 2026.
- Francisco J. García-Pacheco, *Selfadjoint operators on real or complex Banach spaces*, Nonlinear Analysis 192 (2020), Article 111696.

It is a classical result in the Hilbert space context that the approximate spectrum coincides with the spectrum for a normal operator also. The following result provides a partial extension of this property to reflexive smooth Banach spaces.

Theorem 17. *Let X be a reflexive smooth Banach space and $T \in B(X)$. If $\lambda \in \sigma(T)$ is such that $T - \lambda I$ is normal, then $\lambda \in \sigma_{app}(T)$.*

Proof. Consider $\lambda \in \sigma(T) = \sigma_{app}(T) \cup \sigma_{eig}(T')$. If $\lambda \in \sigma_{eig}(T')$, then we obtain a nonzero $f \in X^*$ such that $\|(T' - \lambda I)f\| = 0$. Now, as in the earlier case, since J is onto, there is an $x \neq 0$ in X such that

$$\|(T' - \lambda I)f\| = \|(T - \lambda I)'Jx\| = \|(T - \lambda I)x\|,$$

since $T - \lambda I$ is normal. This proves that $\lambda \in \sigma_{eig}(T) \subseteq \sigma_{app}(T)$. $\square$

Note that in the Hilbert space setting, if T is normal, then $T - \lambda I$ is normal for any scalar λ . As a direct consequence of the above result, we obtain the equality of the minimum modulus and Crawford number for a non-invertible normal operator, generalizing Proposition 3.15 of [7].

Corollary 5. *Let T be a normal operator on a reflexive smooth Banach space X . Then T is invertible if and only if T is bounded below. Hence, if T is a non-invertible normal operator, then $c(T) = \mu(T) = 0$.*

Proof. The first part follows from Theorem 17 by taking $\lambda = 0$, and the second from $0 \leq c(T) \leq \mu(T) = 0$, when T is not invertible. $\square$

Conclusion and open problems

In this article, we have considered three classes of operators on smooth Banach spaces - namely self-adjoint, normal, and unitary operators - and investigated several of their properties, including attainment properties and spectral behaviour. In particular, we obtained a number of results that are analogous to their counterparts in the Hilbert space setting. At the same time, the extent to which other well-known properties from the Hilbert space theory remain valid in the setting of smooth Banach spaces is not yet fully clear and needs further investigation. This leads to several natural questions.

1. Under what conditions does a self-adjoint operator become Hermitian and conversely?
2. While strongly normal operators are known to be self-adjoint and positive, is the converse true, and can one obtain a characterisation of strongly normal operators?
3. If T is a self-adjoint operator on a smooth Banach space, are the operators $\|T\|^2 I - T^2$ and $T^2 - \mu(T)^2 I$ strongly normal, or can counterexamples be found?
4. What can be said about the structure of the numerical range of special operators such as normal, unitary, and self-adjoint operators, and under what conditions is it convex?
5. Is there any relationship between the operator identity $TJ^{-1}T'J = J^{-1}T'JT$ and the norm equality $\|Tx\| = \|T'Jx\|$ for all $x \in X$, which are equivalent in the Hilbert space setting?

Figure 1: Source page containing the first open question.

Hence, we obtain $TJ^{-1}T'J = I$, as required. $\square$

We conclude this section by providing an example of a normal, self-adjoint and unitary operator on $\ell^4(\mathbb{N})$.

Example 2. Consider the operator T on $\ell^4(\mathbb{N})$ defined by

$$T(x) = (x_2, x_1, x_3, x_4, \dots) \text{ for } x = (x_1, x_2, x_3, \dots) \in \ell^4(\mathbb{N}).$$

Consider the transpose T' of this operator defined on $\ell^{4/3}(\mathbb{N})$ by

$$T'(y) = (y_2, y_1, y_3, \dots) \text{ for } y = (y_1, y_2, y_3, \dots) \in \ell^{4/3}(\mathbb{N}).$$

The normalized duality map $J : \ell^4(\mathbb{N}) \rightarrow \ell^{4/3}(\mathbb{N})$ is given as

$$J(x) = \frac{1}{\|x\|_4^2} (|x_1|^2 x_1, |x_2|^2 x_2, |x_3|^2 x_3, \dots)$$

Hence,

$$T'J(x) = \frac{1}{\|x\|_4^2} (|x_2|^2 x_2, |x_1|^2 x_1, |x_3|^2 x_3, \dots) \quad \text{and} \quad J(Tx) = \frac{1}{\|x\|_4^2} (|x_2|^2 x_2, |x_1|^2 x_1, \dots).$$

Thus, $T'J(x) = J(Tx)$, so T is self-adjoint, and hence a normal operator. On calculating the norm, we obtain

$$\|T'J(x)\|_{4/3}^{4/3} = \frac{1}{\|x\|_4^{8/3}} (|x_2|^4 + |x_1|^4 + \dots) = \|x\|_4^{4/3}.$$

Since T is an isometry on $\ell^4(\mathbb{N})$, we get

$$\|T'J(x)\|_{4/3} = \|Tx\|_4 = \|x\|_4 \quad \text{for all } x \in \ell^4(\mathbb{N}),$$

showing that T is a unitary operator.

5 Spectral properties of self-adjoint and normal operators

In this section, we establish some results on the spectrum of self-adjoint operators and normal operators on smooth Banach spaces. In the Hilbert space setting, it is well known that for a self-adjoint operator, the eigenvalues are real, and the eigenvectors corresponding to distinct eigenvalues are orthogonal. In the setting of smooth Banach spaces, it was shown (Theorem 6.3(i)) in [11] that the eigenvalues of self-adjoint operators are real. Here, we obtain an analogous result equivalent to orthogonality of eigenvectors in the setting of smooth Banach spaces.

Proposition 13. Let x_1 and x_2 be eigenvectors corresponding to two distinct eigenvalues of a self-adjoint operator T on smooth Banach space X . Then $x_2 \in {}^\perp J(x_1)$ and $x_1 \in {}^\perp J(x_2)$.

Figure 2: Source page containing the ℓ^4 coordinate-swap example.